

Foundation (Jalapeno)

Q1. Factor the expression $x^2 - 4$

- (A) $(x - 2)(x + 2)$
- (B) $(x - 2)(x - 2)$
- (C) $(x + 2)(x + 2)$
- (D) $x(x - 4)$
- (E) $x(x + 4)$

Q2. Factor the expression $9y^2 - 25$

- (A) $(3y - 5)(3y + 5)$
- (B) $(3y - 5)(3y - 5)$
- (C) $(3y + 5)(3y + 5)$
- (D) $(9y - 25)(y + 1)$
- (E) $(9y + 25)(y - 1)$

Q3. Factor the expression $x^2 - 9$

- (A) $(x - 3)(x + 3)$
- (B) $(x - 3)(x - 3)$
- (C) $(x + 3)(x + 3)$
- (D) $x(x - 9)$
- (E) $x(x + 9)$

Q4. Factor the expression $16x^2 - 49$

- (A) $(4x - 7)(4x + 7)$
- (B) $(4x - 7)(4x - 7)$
- (C) $(4x + 7)(4x + 7)$
- (D) $(16x - 49)(x + 1)$
- (E) $(16x + 49)(x - 1)$

Q5. Factor the expression $25y^2 - 36$

- (A) $(5y - 6)(5y + 6)$
- (B) $(5y - 6)(5y - 6)$
- (C) $(5y + 6)(5y + 6)$
- (D) $(25y - 36)(y + 1)$
- (E) $(25y + 36)(y - 1)$

Q6. Factor the expression $x^2 - 16$

- (A) $(x - 4)(x + 4)$
- (B) $(x - 4)(x - 4)$
- (C) $(x + 4)(x + 4)$
- (D) $x(x - 16)$
- (E) $x(x + 16)$

Q7. Factor the expression $9x^2 - 25$

- (A) $(3x - 5)(3x + 5)$
- (B) $(3x - 5)(3x - 5)$
- (C) $(3x + 5)(3x + 5)$
- (D) $(9x - 25)(x + 1)$
- (E) $(9x + 25)(x - 1)$

Q8. Factor the expression $4x^2 - 9$

- (A) $(2x - 3)(2x + 3)$
- (B) $(2x - 3)(2x - 3)$
- (C) $(2x + 3)(2x + 3)$
- (D) $(4x - 9)(x + 1)$
- (E) $(4x + 9)(x - 1)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2 - 64$ and explain your steps

Q10. Factor the expression $49y^2 - 25$ and provide a real-world scenario where this expression would be used

Chef's Tips

- Tip 1: Emphasize the importance of recognizing the difference of squares pattern
- Tip 2: Provide examples of real-world scenarios where factoring is applied
- Tip 3: Encourage students to check their work by multiplying the factors